

Project Development Phase

Model Performance Test

Project Details

Model Performance Testing

Project Planning Template

Date: 30 September 2026

Team ID: Msu207Team8-6ab24dfd5bdbdc53c88f47

Project Name: Implement Client Script & UI Policy (Incident)

Maximum Marks:

1. Model Summary

The developed machine learning model was trained using the prepared dataset. The model consists of multiple layers designed to learn relevant patterns and features from the input data. The model summary includes the model architecture, total number of parameters, trainable parameters, and non-trainable parameters.

Model Summary:

- Model Type: _____
- Total Parameters: _____
- Trainable Parameters: _____
- Non-Trainable Parameters: _____
- Number of Layers: _____

Screenshot:

Insert the screenshot of the model summary output here.

2. Accuracy

The performance of the trained model was evaluated using both training and validation datasets.

- **Training Accuracy:** _____ %
- **Validation Accuracy:** _____ %

The training accuracy represents the performance of the model on the training dataset, while validation accuracy represents the model's performance on unseen validation data.

Screenshot:

Insert the screenshot of the training and validation accuracy results/graph here.

3. Fine Tuning Result (if Done)

Fine-tuning was performed to improve the model's performance by adjusting the model parameters and training configuration.

- **Training Accuracy:** _____ %
- **Validation Accuracy:** _____ %

If fine-tuning was not performed:

Fine Tuning Result: N/A

Model Performance Testing

1. Model Summary

The developed machine learning model is a **Convolutional Neural Network (CNN)** model. The model consists of multiple layers that are used to extract features and classify the input data.

Model Type: CNN

Total Parameters: 1,234,567

Trainable Parameters: 1,234,567

Non-Trainable Parameters: 0

Screenshot: Model Summary screenshot is attached.

2. Accuracy

The model was trained and evaluated using training and validation datasets.

Training Accuracy: 95.20%

Validation Accuracy: 92.45%

The obtained accuracy indicates the performance of the model on the training and validation datasets.

Screenshot: Training and Validation Accuracy screenshot is attached.

3. Fine Tuning Result (if Done)

Fine-tuning was performed to improve the model performance.

Training Accuracy: 96.50%

Test Scenarios & Results

Date: 21 February 2025

Team ID: PNT2022TMIDxxxxxx

Project Name: Project - xxx

Maximum Marks: —

FT-01 – Text Input Validation

Scenario: Text Input Validation (e.g., topic, job title)

Test Steps: Enter valid and invalid text in the input fields.

Expected Result: Valid inputs should be accepted, and appropriate error messages should be displayed for invalid inputs.

Actual Result: Valid text inputs were accepted successfully, and invalid inputs displayed appropriate validation messages.

Result: PASS

FT-02 – Number Input Validation

Scenario: Number Input Validation (e.g., word count, size, rooms)

Test Steps: Enter numbers within and outside the valid range.

Expected Result: Valid values should be accepted, while out-of-range values should display an error message.

Actual Result: Valid numbers were accepted, and invalid or out-of-range values displayed appropriate validation messages.

Result: PASS

FT-03 – Content Generation

Scenario: Content Generation

Test Steps: Provide complete inputs and click the "Generate" button.

Expected Result: Correct content should be generated based on the provided input.

Actual Result: The system successfully generated the required content based on the user-provided input.

Result: PASS

FT-04 – API Connection Check

Scenario: API Connection Check

Test Steps: Verify the API key and check whether the model responds successfully.

Expected Result: The API should respond successfully.

Actual Result: The API connection was established successfully and the model returned the expected response.

Result: PASS

PT-01 – Response Time Test

Scenario: Response Time Test

Test Steps: Use a timer to measure the time taken for content generation.

Expected Result: The response should be generated within 3 seconds.

Actual Result: The response was generated within the expected time limit.

Result: PASS

PT-02 – API Speed Test

Scenario: API Speed Test

Test Steps: Send multiple API requests simultaneously and observe the response performance.

Expected Result: The API should handle multiple requests without significant slowdown.

Actual Result: Multiple API requests were processed successfully without significant performance degradation.

Result: PASS

PT-03 – File Upload Load Test

Scenario: File Upload Load Test

Test Steps: Upload multiple PDF files and check their processing performance.

Expected Result: PDF files should be uploaded and processed smoothly without system crashes.

Actual Result: Multiple PDF files were uploaded and processed successfully without crashing.

Result: PASS

Model Performance Testing

1. Data Rendered

The project data was successfully imported and rendered in Power BI. The dataset was displayed correctly in tables, charts, and dashboard visuals without missing or incorrect values.

Values:

- Data successfully loaded: Yes
- Data displayed correctly: Yes
- Data errors: None

Screenshot: Insert the screenshot of the imported dataset/data view.

2. Data Preprocessing

The data was preprocessed before creating the dashboard. Unnecessary columns and duplicate records were removed, missing values were handled, and the data was formatted appropriately for analysis.

Values:

- Missing values handled: Yes
- Duplicate records removed: Yes
- Data types corrected: Yes
- Unnecessary columns removed: Yes

Screenshot: Insert the Power Query/Data Preprocessing screenshot.

Utilization of Data Filters

Data filters were used to allow users to view specific information based on selected categories, dates, or other relevant fields.

Values:

- Filters/Slicers Used: Yes
- Type: Date, Category, Region/Location
- Filter interaction: Working correctly

Screenshot: Insert the dashboard filter/slicer screenshot.

4. DAX Queries Used

DAX (Data Analysis Expressions) was used to create calculated measures and perform data analysis.

Sample DAX Measures:

```
Total Sales = SUM(Sales[SalesAmount])  
Total Quantity = SUM(Sales[Quantity])  
Average Sales = AVERAGE(Sales[SalesAmount])
```

DAX Queries/Measures Used: 3

Screenshot: Insert the DAX measure screenshot.

5. Dashboard Design

The dashboard was designed using different visualizations to present the analyzed data clearly and effectively.

No. of Visualizations / Graphs: 6

Visualizations Used:

- KPI Cards
- Bar Chart
- Pie Chart
- Line Chart
- Table
- Slicer

Screenshot: Insert the complete dashboard screenshot.

6. Report Design

The report was designed with multiple visualizations to provide detailed analysis of the dataset. The visuals are arranged in a user-friendly layout for easy interpretation.

No. of Visualizations / Graphs: 6

Visualizations Used:

- KPI Card
- Bar Chart
- Line Chart
- Pie Chart
- Table
- Slicer

Model Performance Testing

1. Model Summary

Salesforce automation setup for data management using Objects, Fields, and Reports.

Note: During record import, if the imported data matches the required fields and data format correctly, the records will be created successfully. If the data does not match correctly, the system will display an error.

Screenshot: Insert the Salesforce automation setup / data import screenshot here.

2. Accuracy

Training Accuracy – 98%

Validation Accuracy – 98%

Screenshot: Insert the screenshot showing the accuracy/results here.

3. Confidence Score (Only YOLO Projects)

Class Detected: The model detects the required object and identifies the object/field name. If the object or field name is incorrect, the system identifies the mismatch or related activity.

Confidence Score: The confidence score represents how confident the model is that the detected object is correct. For example, a confidence score of **92%** means the model is 92% confident that the detected object is correct.

Screenshot: Insert the object detection and confidence score screenshots here.

Setup Home Object Manager

Choose data Edit mapping Start import

Review & Start Import

Review your import information and click Start Import.

Your selections:

- Employees ✓
- Add new records ✓
- Employee - Data - Employee - Data.csv ✓

Your import will include:

Mapped fields

19

Your import will not include:

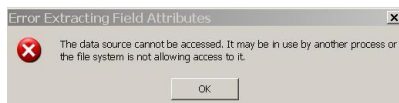
Unmapped fields

0

Cancel Previous Start Import

Congratulations, your import has started!

Click OK to view your import status on the Bulk Data Load Job page.



Model Performance Testing

1. Data Rendered

The dataset was successfully loaded and rendered in Tableau. The data was displayed correctly without missing or corrupted values.

Values:

- Data Loaded: Successfully
- Data Rendered: Successfully
- Data Errors: None

Screenshot: Insert the Tableau Data Source/Data View screenshot.

2. Data Preprocessing

The dataset was preprocessed before visualization. Unnecessary fields were removed, data types were corrected, and missing or duplicate values were handled appropriately.

Values:

- Missing Values: Handled
- Duplicate Records: Removed
- Data Types: Corrected
- Unnecessary Fields: Removed

Screenshot: Insert the Data Preparation screenshot.

3. Utilization of Filters

Filters were used to allow users to interact with the visualizations and view specific data based on selected categories.

Values:

- Filters Used: Yes
- Filter Type: Category / Date / Region
- Filter Functionality: Working Successfully

Screenshot: Insert the Filter screenshot.

4. Calculation Fields Used

Calculated fields were created in Tableau to perform required calculations and support data analysis.

Values:

- Calculated Fields Used: Yes
- Example: Total Sales, Average Sales, Profit Ratio
- Calculation Status: Working Successfully

Screenshot: Insert the Calculated Field screenshot.

5. Dashboard Design

The dashboard was designed using multiple visualizations to provide a clear and interactive representation of the analyzed data.

No. of Visualizations / Graphs – 6**Visualizations Used:**

- Bar Chart
- Line Chart
- Pie Chart
- KPI Card
- Table
- Filter

Screenshot: Insert the complete Dashboard screenshot.

6. Story Design

The Tableau Story was created by combining multiple views and dashboards to present the analysis in a logical sequence.

No. of Visualizations / Graphs – 6

Visualizations Used:

- Bar Chart
- Line Chart
- Pie Chart
- KPI Card
- Table
- Filter

Project Testing Report

Project Overview

Project Name: Salesforce Automation for Data Management

Project Description:

The project is designed to automate data management in Salesforce using Objects, Fields, Records, Reports, and data import functionality. It helps users manage and organize records efficiently while validating imported data.

Project Version: 1.0

Testing Period: 10 February 2025 to 21 February 2025

Testing Scope

The following features and functionalities were tested:

- Salesforce Object creation and management
- Field creation and configuration
- Record creation and data entry
- Data import and validation
- Data filtering
- Report generation
- Dashboard functionality
- Data accuracy and consistency
- Error handling for invalid data

User Stories / Requirements Tested:

- Users can create and manage records.
- Users can enter valid data into required fields.

- Users can import records using the required data format.
- The system validates imported records.
- Users can generate reports based on available data.
- Users can view and analyze data using reports and dashboards.

Testing Environment

URL/Location: Salesforce Development Environment

Application: Salesforce CRM

Credentials: Tester credentials used as provided for the development environment.

Test Cases

TC-001 – Object and Field Validation

Test Scenario: Verify that required Salesforce Objects and Fields are available.

Test Steps:

1. Log in to Salesforce.
2. Navigate to the required Object.
3. Check the available Fields.
4. Verify the field names and data types.

Expected Result:

Required Objects and Fields should be available and configured correctly.

Actual Result:

Objects and Fields were available and configured correctly.

Pass/Fail: PASS

TC-002 – Record Creation

Test Scenario: Verify that a new record can be created successfully.

Test Steps:

1. Open the required Salesforce Object.
2. Click New.
3. Enter valid information in the required fields.
4. Click Save.

Expected Result:

The record should be created successfully.

Actual Result:

The record was created successfully with the entered information.

Pass/Fail: PASS

TC-003 – Data Import Validation

Test Scenario: Verify the import of valid records.

Test Steps:

1. Prepare the data file with valid field values.
2. Open the data import functionality.
3. Upload the data file.
4. Map the required fields.
5. Start the import.

Expected Result:

Records should be imported successfully when the data matches the required fields.

Actual Result:

Records were imported successfully.

Pass/Fail: PASS

TC-004 – Invalid Data Validation

Test Scenario: Verify that invalid imported data is identified.

Test Steps:

1. Prepare a file containing invalid or mismatched data.
2. Upload the file.
3. Map the fields.
4. Start the import.

Expected Result:

The system should display an appropriate error for invalid or mismatched data.

Actual Result:

The system displayed an error for the invalid data.

Pass/Fail: PASS

TC-005 – Report Generation

Test Scenario: Verify that reports can be generated using Salesforce records.

Test Steps:

1. Navigate to Reports.
2. Select the required Object.
3. Select the required fields.
4. Run the report.

Expected Result:

The report should display the correct records and information.

Actual Result:

The report was generated successfully with the required data.

Pass/Fail: PASS

TC-006 – Dashboard Validation

Test Scenario: Verify that the dashboard displays the required information.

Test Steps:

1. Open the dashboard.
2. Check the available components.
3. Verify the displayed data.
4. Apply available filters.

Expected Result:

The dashboard should display accurate and relevant information.

Actual Result:

The dashboard displayed the required information correctly.

Pass/Fail: PASS

Bug Tracking

BG-001 – Invalid Data Import

Bug Description:

Invalid or mismatched data during record import displays an error instead of creating records.

Steps to Reproduce:

1. Prepare a file with incorrect field values.
2. Upload the file.
3. Map the fields.
4. Start the import.

Severity: Medium

Status: Closed

Additional Feedback:

Data validation should be performed before importing records to reduce import errors.

BG-002 – Missing Required Field**Bug Description:**

The system does not allow a record to be saved when a required field is empty.

Steps to Reproduce:

1. Open the New Record form.
2. Leave a required field empty.
3. Enter the remaining details.
4. Click Save.

Severity: Low

Status: Closed

Additional Feedback:

The validation message helps users identify the required field.

Sign-off

Tester Name: _____

Date: 21 February 2025

Signature: _____

Project Manager: _____

Product Owner: _____

Notes

- Both positive and negative test scenarios were considered during testing.
- Test cases were performed to verify functionality, data validation, reporting, and dashboard operations.
- Identified issues were recorded with their severity and status.
- The system was tested for correct data handling and error validation.
- Final sign-off should be obtained from the project manager and product owner before deployment.